AI as a Capability-Contingent Amplifier of Knowledge Worker Performance

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Abstract

Generative artificial intelligence (AI) is rapidly transforming knowledge work, yet its effects on individual productivity remain uneven and poorly understood. Drawing on theories of task–technology fit and human capital complementarities, this study examines how AI usage interacts with task structure and worker capability to shape performance outcomes. Using a preregistered survey of knowledge workers, I estimate ordinal logistic regression models predicting self-reported formal evaluation outcomes.

Results show that AI usage is positively associated with performance on average, but with substantial heterogeneity in marginal returns. In particular, the relationship between AI intensity and performance is stronger among workers with higher digital capability and under more structured task conditions. These findings suggest that AI functions as a capability-contingent amplifier, producing uneven returns depending on the alignment between technology use, worker capability, and task structure.

Generative AI has rapidly entered the workflow of knowledge workers, raising fundamental questions about whether and under what conditions AI improves performance. A growing empirical literature shows that AI assistance can produce substantial gains in productivity and quality in specific task environments. For example, experimental evidence shows that access to generative AI can significantly reduce completion time and improve output quality in writing tasks (Noy & Zhang, 2023), while field evidence from customer support settings shows meaningful productivity gains, particularly among less-experienced workers (Brynjolfsson et al., 2025). At the same time, other studies document important limitations: AI can degrade performance when tasks fall outside the system's capability frontier or when users misapply its outputs (Dell'Acqua et al., 2026; Fitzpatrick et al., 2025).

Taken together, these findings point to a central but underdeveloped question in organizational research: why does AI improve performance in some settings but not others? Existing work has largely focused on estimating average treatment effects within specific task environments, leaving the sources of heterogeneity in AI effectiveness less well understood. Understanding this heterogeneity is not only theoretically important but also central to how organizations deploy AI tools, as uneven returns across workers and tasks may shape the effectiveness of AI adoption strategies.

This paper argues that the performance effects of AI are fundamentally contingent on the interaction between technology, worker capability, and task conditions. This perspective builds on the concept of complementarities in organizational economics, where the returns to one input depend on the presence of others (Milgrom & Roberts, 1990). Rather than operating as a uniform productivity enhancer, AI functions as a capability-contingent amplifier that can either enhance or undermine performance depending on how it is used.

A key mechanism underlying this perspective is calibration—the extent to which individuals accurately assess both their own capabilities and the reliability of AI outputs. Recent research shows that even when AI systems are highly accurate, performance gains depend on whether users are able to appropriately rely on them, with miscalibrated users failing to benefit from AI assistance (Caplin et al., 2024). More broadly, work on human–AI complementarity suggests that algorithmic systems do not replace human judgment but instead interact with it, such that the returns to AI depend on how human expertise shapes both the use and interpretation of algorithmic outputs (Allen & Choudhury, 2020).

Building on these insights, I develop a framework in which the effects of AI usage on performance depend on complementarities between AI usage, worker capability, and task structure. In the complementarities framework, practices or inputs are mutually reinforcing, such that the marginal return to one factor increases in the presence of others (Milgrom & Roberts, 1990). Applying this logic to AI, the effectiveness of AI usage is expected to depend not only on the technology itself but also on its alignment with worker capability and task conditions. This perspective implies that organizations are unlikely to realize uniform returns from AI adoption, and that the value of AI may depend on how it is combined with investments in worker capability and task design.

The analysis is based on a preregistered survey of knowledge workers (N = 102). Because the data are cross-sectional and self-reported, the analyses are associational rather than causal. The goal is therefore not to estimate definitive causal effects but to develop and provide initial evidence for a theory of capability-contingent AI performance.

This paper contributes to the emerging literature on AI and work in three ways. First, it extends contingency-based perspectives on work design by showing that the performance effects of AI depend jointly on worker capability and task structure rather than on technology use alone. Second, it introduces perceived digital capability as a theoretically relevant boundary condition linking self-efficacy and human–AI interaction. Third, it develops a complementarity perspective in which AI functions as a capability-contingent amplifier, such that the returns to AI usage depend on alignment between technology use, worker capability, and task structure. In doing so, the paper offers an initial theoretical foundation for explaining why organizations may observe uneven returns to AI adoption across workers and work contexts.

THEORY AND HYPOTHESES

Generative AI and heterogeneous performance effects

Generative AI systems have the potential to reshape knowledge work by reducing search costs, accelerating drafting, and supporting complex cognitive tasks through context-aware suggestions and partial automation (Cui et al., 2026; Peng et al., 2023). However, emerging evidence suggests that the productivity effects of AI are inherently heterogeneous and depend on how humans interact with algorithmic systems rather than on the technology alone.

A growing body of research shows that AI does not uniformly improve performance but instead changes the distribution of outcomes across workers. For example, AI assistance has been shown to improve average performance while disproportionately benefiting lower-performing individuals, thereby reducing performance inequality (Brynjolfsson et al., 2025). At the same time, AI can degrade performance when tasks fall outside the system's domain of competence or when users misapply its outputs (Dell'Acqua et al., 2026).

Recent experimental evidence further highlights the role of user capability and judgment in shaping AI effectiveness. Performance gains from AI depend not only on baseline ability but also on calibration—how accurately individuals assess their own competence and the reliability of AI outputs. Well-calibrated individuals benefit more from AI assistance, while miscalibrated users fail to fully realize its potential, even when the technology itself is highly accurate (Caplin et al., 2024). These findings suggest that AI effectiveness is fundamentally shaped by the interaction between human capability and technology use.

A growing empirical literature provides evidence of such heterogeneity. In a field setting, AI assistance produced larger productivity gains among less-experienced workers, suggesting that AI may substitute for certain forms of human capital (Brynjolfsson et al., 2025). In contrast, other work shows that AI can both improve and degrade performance depending on task characteristics, particularly when tasks fall outside the system's domain of competence (Dell'Acqua et al., 2026). Together, these findings suggest that the effects of AI are inherently contingent and that understanding its impact requires attention to the interaction between technology, workers, and tasks.

These patterns are consistent with a broader literature on human–AI complementarity, which emphasizes that algorithmic systems do not replace human judgment but instead interact with it. Prior work shows that human expertise is critical for interpreting, validating, and appropriately integrating algorithmic outputs, particularly in complex or uncertain environments. At the same time, expertise can introduce frictions, such as algorithm aversion, which may limit the extent to which individuals rely on AI recommendations (Allen & Choudhury, 2020). As a result, the performance effects of AI are not monotonic in worker capability but depend on how capability shapes both the ability and willingness to use AI effectively.

Building on this perspective, I develop a framework in which the performance effects of AI depend on complementarities between AI usage, worker capability, and task structure.

Hypotheses

Task structure as a moderator

The extent to which AI improves performance is likely to depend on the structure of the task environment. When tasks are highly standardized—characterized by clear procedures, well-defined goals, and objective evaluation criteria—AI-generated outputs are easier to interpret, verify, and integrate into existing workflows. Under such conditions, AI can function as a reliable augmentation tool that enhances efficiency without substantially increasing cognitive load.

By contrast, when tasks are less structured and require subjective judgment or problem framing, AI outputs may be more difficult to evaluate and apply. Ambiguity in task requirements increases the risk of misalignment between AI-generated suggestions and task objectives, potentially reducing or even reversing performance gains.

H1. The relationship between AI usage intensity and performance is more positive when task structure is high.

Worker capability and complementarity

In addition to task conditions, the effectiveness of AI depends on the capabilities of the worker using it. Digital capability reflects individuals' perceived ability to effectively use digital tools, consistent with prior research on computer self-efficacy (Compeau & Higgins, 1995). Beyond technical proficiency, effective AI use requires the ability to evaluate the reliability of outputs, adapt them to context, and integrate them into ongoing workflows.

Prior research suggests that human capability plays a central role in determining whether algorithmic tools enhance or hinder performance. Individuals with greater capability are better able to detect when AI outputs are incorrect and to selectively rely on AI when it is beneficial. At the same time, capability also shapes reliance behavior, as more capable individuals may underutilize AI due to overconfidence or skepticism toward algorithmic advice (Allen & Choudhury, 2020; Caplin et al., 2024). This dual role implies that the relationship between AI usage and performance is contingent on both the level and the behavioral implications of worker capability.

These capabilities enable workers to extract greater value from AI systems and to mitigate potential errors or limitations in AI-generated content. From a complementarity perspective, the marginal returns to AI usage are expected to increase with worker capability, as more capable individuals are better positioned to integrate AI into their workflows and to translate AI-generated outputs into performance improvements (Milgrom & Roberts, 1990).

H2. Digital capability moderates the relationship between AI usage intensity and performance, such that the relationship is more positive at higher levels of capability.

Configurational alignment of technology, task, and capability

Taken together, these perspectives suggest that AI effectiveness depends on the joint alignment of technology, task, and worker characteristics. This configurational perspective is consistent with complementarities theory, which emphasizes that performance gains arise from coherent systems of mutually reinforcing elements rather than from any single factor in isolation (Milgrom & Roberts, 1990). AI is most likely to enhance performance when tasks are sufficiently structured to allow outputs to be evaluated and when workers possess the capability to appropriately interpret and integrate those outputs.

In contrast, misalignment along either dimension—low capability or low task structure—may limit the benefits of AI or even produce negative performance effects. This suggests a configurational perspective in which the productivity effects of AI emerge from the interaction of multiple complementary factors rather than from any single dimension in isolation.

H3. The positive relationship between AI usage intensity and performance is strongest when both task structure and digital capability are high.

METHODS

Study design and sample

The data come from a pilot survey of knowledge workers recruited through an online panel. After data-quality filtering, the full analytic sample includes 102 respondents. Because the formal evaluation

outcome is only observed for respondents who reported receiving a recent formal evaluation, the primary analyses reported in this paper are conducted on a subsample of 69 observations. The study was preregistered prior to data collection. The preregistration specified hypotheses and analysis plans focused on the relationship between AI usage and performance. The hypotheses developed in this paper extend the scope of the original preregistration and are informed by patterns observed in the pilot data. Accordingly, the analyses should be interpreted as theory-building rather than strictly confirmatory tests. The original preregistration is available at https://osf.io/6gqat/overview.The focal outcome is respondents' self-reported formal evaluation outcome, which captures perceived standing in organizational performance appraisal systems rather than an independently verified objective performance record.

Measures

AI usage intensity is operationalized as a composite index capturing the extent to which respondents use AI tools in their daily work. The index combines three components: (1) frequency of AI tool use, (2) percentage of work time spent using AI, and (3) degree of AI integration into workflow. Each component is standardized prior to aggregation, and the index is computed as the average of the standardized items. The resulting scale exhibits high internal consistency (Cronbach's $\alpha = 0.87$).

Task standardization is measured using a multi-item index capturing the extent to which work tasks are structured and rule-based. The index includes items assessing task clarity, clarity of performance criteria, and solution clarity, as well as an item capturing the extent to which tasks require subjective judgment (reverse-coded). Items are averaged on their original 1–5 scale to form the index. The scale demonstrates good internal reliability (Cronbach's $\alpha = 0.80$).

Digital capability reflects respondents' perceived ability to effectively use digital tools in their work, consistent with prior research on computer self-efficacy (Compeau & Higgins, 1995). The measure is constructed from self-reported assessments and standardized prior to inclusion in regression models.

All continuous predictors are mean-centered (z-standardized) prior to estimation to facilitate interpretation of interaction terms. Control variables include tenure, experience, and hours worked per week.

Measurement considerations

Because the focal constructs are measured using self-reports collected in a single survey, common method variance is a potential concern. This concern is particularly relevant for the observed associations between AI usage, perceived digital capability, and performance-related outcomes. At the same time, the focal claims in the paper center on interaction patterns rather than on large uniform main effects, and such differential relationships are less likely to be explained entirely by a single common response tendency. Nevertheless, the results should be interpreted with appropriate caution, and future research should examine these relationships using multi-source measures of performance and technology use.

Empirical strategy

The analyses use ordinal logistic regression models to examine whether the relationship between AI usage intensity and self-reported formal evaluation outcomes varies with task structure and digital capability. The empirical strategy proceeds in stages, beginning with baseline associations and then estimating pairwise interaction models corresponding to Hypotheses 1 and 2. Given the pilot sample size and the limited number of observations with formal evaluation outcomes, the proposed three-way relationship in Hypothesis 3 is not estimated directly. Instead, it is evaluated indirectly through the joint interpretation of the pairwise interaction patterns and their predicted probabilities.

RESULTS

Descriptive Statistics

Table 1 reports descriptive statistics and correlations among the main variables. The distribution of the performance outcome indicates that most respondents report moderate to high evaluations, with fewer observations at the extreme upper end. This distribution motivates focusing on the probability of achieving high performance (ratings of 4 or 5) when interpreting model predictions.

Correlations are generally modest in magnitude, suggesting that multicollinearity is unlikely to pose a major concern in the regression analyses.

Table 1 Descriptive Statistics, Reliabilities, and Correlations

	Mean	S.D.	1	2	3	4	5	6	7	8
1. AI intensity index	0.00	0.89	0.30							
2. Task standardization index	3.72	0.78	-0.08	0.80						
3. Digital capability (digital comfort)	4.29	0.75	0.22	-0.01	—					
4. Formal evaluation outcome	3.84	0.63	0.06	0.11	0.31	—				
5. Relative performance	3.73	0.72	0.27	0.08	0.28	0.50	—			
6. Tenure (years)	6.50	5.66	-0.22	0.08	-0.12	0.00	-0.11	—		
7. Experience (years)	18.84	11.70	-0.19	0.11	-0.17	0.02	-0.09	0.48	—	
8. Hours worked per week	41.54	4.71	0.14	-0.04	0.04	0.25	0.09	0.16	0.15	—

Notes. Cronbach's alpha coefficients are shown on the diagonal where applicable. Correlations are pairwise Pearson correlations shown below the diagonal. --- indicates a single-item measure or reliability not applicable.

Main effects and interaction models

Table 2 presents the results of ordinal logistic regression models predicting self-reported formal evaluation outcomes. All regression models in Table 2 are estimated on the subsample with observed formal evaluation outcomes (N = 69). Across specifications, the main effect of AI usage is not uniformly large, but interaction terms indicate that its relationship with performance depends on both worker capability and task structure.

In particular, the interaction between AI intensity and digital capability is positive, indicating that the association between AI usage and higher performance evaluations is stronger among workers

with greater perceived digital capability. Similarly, interactions involving task standardization suggest that the effectiveness of AI usage varies across task environments.

Because coefficients from ordinal logistic models are not directly interpretable in probability terms, I next examine predicted probabilities to provide a more substantively meaningful interpretation of these interaction effects.

Table 2 Ordinal Logistic Regression Models Predicting Formal Evaluation Outcome

	(1)	(2)	(3)
AI intensity	1.011	1.015	9.320000e-01
	(0.345)	(0.368)	(0.339)
Task standardization		1.424	
		(0.486)	
Digital capability			2.451000e+00*
			(0.837)
AI intensity × Task standardization		1.721	
		(0.512)	
AI intensity × Digital capability			1.908000e+00
			(0.622)
Tenure	1.023	1.047	1.013000e+00
	(0.060)	(0.065)	(0.061)
Hours worked	1.144	1.085	1.169000e+00
	(0.105)	(0.102)	(0.121)
Job level: 2	8.883	12.764*	6.237000e+00
	(10.432)	(15.437)	(7.660)
Job level: 3	1.689	1.303	1.260000e+00
	(1.272)	(1.037)	(1.025)
Job level: 4	22.074	27.544	1.334800e+01
	(35.243)	(45.932)	(21.533)
Job level: 6	24977996.113***	35313542.840***	1.075523e+08***
	(243.780)	(378.553)	(652.955)
Job function: 2	0.525	0.558	5.560000e-01
	(0.559)	(0.607)	(0.647)
Job function: 3	31.753*	69.019**	2.835800e+01*
	(43.914)	(99.184)	(40.734)
Job function: 4	0.062	0.097	1.170000e-01
	(0.117)	(0.190)	(0.218)
Job function: 5	2.850	1.806	8.512000e+00
	(5.385)	(3.399)	(18.647)
Job function: 6	0.673	0.642	3.488000e+00
	(0.727)	(0.704)	(4.753)
Job function: 7	4.871	2.053	2.427000e+00
	(14.489)	(6.544)	(7.747)
Job function: 8	5.667	5.395	6.598000e+00
	(7.357)	(7.149)	(9.284)
Job function: 9	1.694	3.418	1.972000e+00
	(3.226)	(6.919)	(3.410)
Job function: 10	1.754	3.030	1.362000e+00
	(2.153)	(3.925)	(1.751)
Job function: 11	0.976	0.947	1.119000e+00
	(0.811)	(0.800)	(0.967)
N	69	69	69

Notes. Entries are odds ratios from ordinal logistic regression models; standard errors are shown in parentheses. Model (1) reports the baseline specification, Model (2) adds the preregistered interaction between AI intensity and task standardization, and Model (3) adds the exploratory interaction between AI intensity and digital capability. Threshold parameters are omitted for readability. * $p < .05$, ** $p < .01$, *** $p < .001$.

Capability-contingent and configurational patterns

Because coefficients from ordinal logistic models are not directly interpretable in probability terms, I examine predicted probabilities to provide a more substantively meaningful interpretation of the interaction effects. In the figures below, performance is defined as the predicted probability of achieving a high formal evaluation (ratings of 4 or 5).

Figure 1 shows the predicted probability of high performance as a function of AI usage across levels of task structure, holding other variables constant.

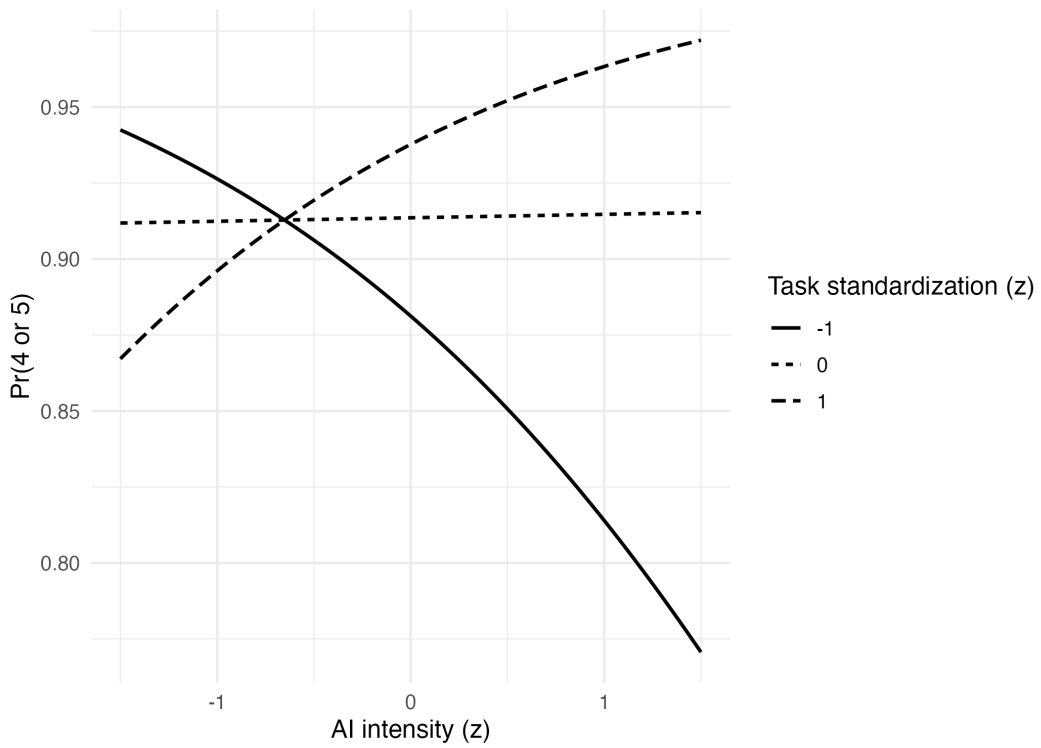

Figure 1: Predicted probability of high performance as a function of AI intensity and task standardization.

Consistent with Hypothesis 1, task structure moderates the relationship between AI usage and performance. Under more standardized task conditions, the relationship between AI usage and performance is more stable and predictable, with increases in AI intensity associated with steady improvements in the probability of high performance. By contrast, under less structured task conditions, the relationship is more variable, and gains from AI usage appear less consistent. These results provide support for H1.

Figure 2 shows the predicted probability of high performance as a function of AI usage across levels of digital capability.

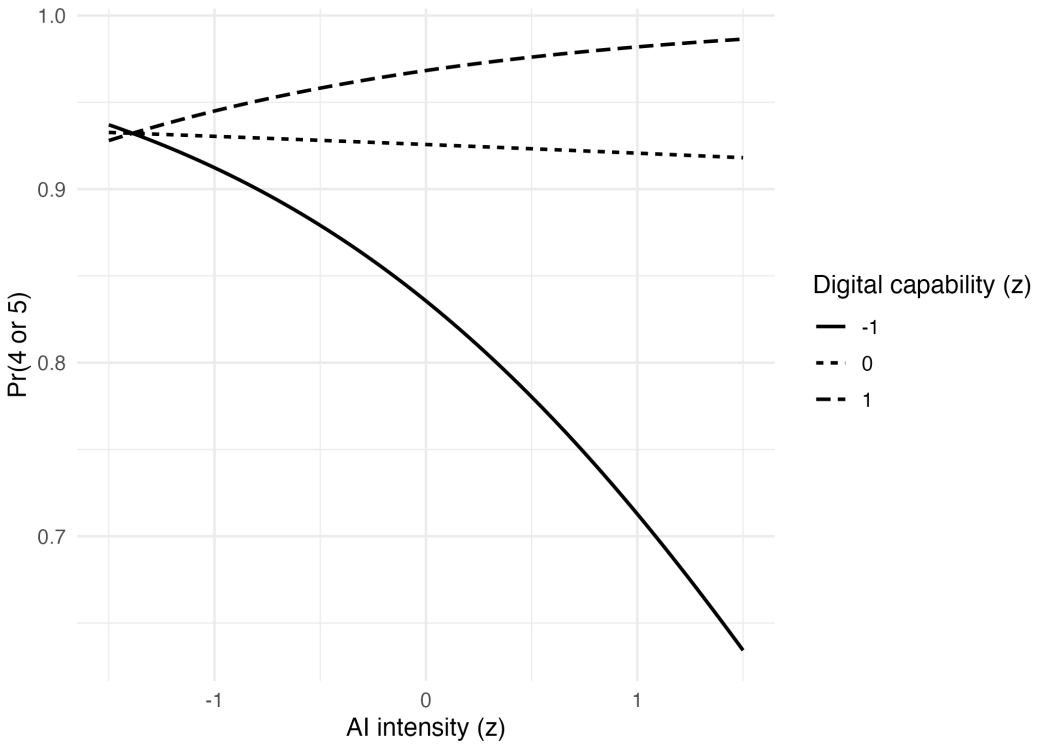

Figure 2: Predicted probability of high performance as a function of AI intensity and digital capability.

Consistent with Hypothesis 2, the relationship between AI usage and performance varies systematically with digital capability. While AI usage is generally associated with a high probability of strong performance across all levels of capability, the magnitude of this relationship differs: the slope of the relationship between AI usage and performance is steeper for workers with higher digital capability. Thus, while AI usage appears broadly beneficial, its marginal returns are substantially larger for workers who are better able to effectively use and interpret digital tools. These results provide support for H2.

Taken together, these patterns suggest that the productivity effects of AI are contingent rather than uniform, and vary in a graded manner across workers and task conditions. Rather than indicating that AI is beneficial only under narrowly defined conditions, the figures suggest that AI usage is generally associated with stronger performance, while the magnitude and stability of those gains depend on the alignment between worker capability and task structure. Although the pilot sample does not support stable estimation of a full three-way interaction, the observed pairwise patterns are consistent with a broader configurational perspective in which the returns to AI depend on complementarities between technology use, human capability, and task structure.

DISCUSSION

This study provides initial evidence consistent with the view that the relationship between generative AI use and worker performance is contingent on both individual and task characteristics, and varies in a graded manner across contexts. When examining the probability of achieving high performance,

AI usage is generally associated with improved performance across conditions, but the magnitude of these improvements varies systematically across workers and tasks.

These patterns are consistent with a capability-contingent view of AI as an amplifier of human capital. Workers with higher perceived digital capability derive greater performance gains from increased AI usage, suggesting that AI complements, rather than substitutes for, human expertise. This finding aligns with prior evidence that the productivity effects of AI are unevenly distributed across workers (e.g., Brynjolfsson et al. (2025)).

More broadly, the results highlight the role of calibration in shaping the returns to AI usage. Even when AI systems are capable, performance gains depend on whether users can appropriately interpret and integrate AI-generated outputs. Prior work shows that miscalibrated reliance on AI can limit performance gains (Caplin et al., 2024). The present findings extend this perspective by suggesting that calibration operates along a continuum, influencing the magnitude—rather than the direction—of AI's impact on performance.

Task structure further conditions these relationships. Under more standardized tasks, the relationship between AI usage and performance is more stable, whereas under less structured tasks, performance gains are more variable and more strongly dependent on worker capability. This is consistent with prior research showing that AI effectiveness depends on task characteristics and the alignment between human judgment and algorithmic output (e.g., Dell'Acqua et al. (2026)).

The primary theoretical contribution of this study is to integrate worker capability and task structure into a unified complementarity framework. Building on the tradition of complementarities in organizational economics (Milgrom & Roberts, 1990), the paper suggests that the returns to AI usage depend on the joint presence of aligned worker, task, and technology conditions. Prior research has documented heterogeneity in the effects of AI across workers and tasks (e.g., Brynjolfsson et al. (2025); Dell'Acqua et al. (2026)), but has largely examined these dimensions in isolation. The present study offers initial evidence that these dimensions may jointly shape the marginal returns to AI usage, thereby motivating future confirmatory research using larger and more behaviorally grounded datasets.

Methodologically, the study adopts a staged empirical approach that prioritizes statistically stable estimation of pairwise complementarities while avoiding overinterpretation of higher-order interactions that are not well supported by the data. This approach is particularly appropriate in pilot settings, where sample size constraints can render complex interaction estimates unstable and difficult to interpret.

PRACTICAL IMPLICATIONS AND LIMITATIONS

These findings contribute to the literature on AI and work by shifting the focus from whether AI improves performance to how its benefits are distributed across workers and tasks. While prior research has emphasized average treatment effects, the results highlight the importance of understanding heterogeneity in marginal returns to AI usage.

The results support a complementarity-based view of human–AI interaction, but suggest that this complementarity operates along a continuum rather than through sharp thresholds. AI does not simply enhance or undermine performance; instead, it produces varying levels of performance gains depending on user capability and task conditions. This perspective extends prior work on human–AI complementarity by emphasizing the role of graded capability differences in shaping outcomes.

The findings also contribute to research on digital capability and self-efficacy by demonstrating that perceived capability is not only associated with performance directly, but also moderates the effectiveness of AI usage. This suggests that digital capability plays a central role in determining how individuals translate access to AI into performance improvements.

Several limitations qualify these findings. First, the data are cross-sectional and based on self-reports, which limits causal inference and raises the possibility of common method inflation. Second, the focal models rely on a subsample of respondents with observed formal evaluation outcomes ($N = 69$), which constrains statistical power and makes higher-order interactions difficult to estimate reliably. Third, although the survey itself was preregistered, the hypotheses developed in this paper extend beyond the original preregistration and should therefore be interpreted as theory-building rather than strictly confirmatory tests. These features do not invalidate the observed patterns, but they do imply that the current study should be viewed as an initial empirical step rather than a definitive test of the proposed framework.

For organizations, the findings suggest that investments in AI tools are unlikely to yield uniform returns across the workforce. Instead, the value of AI adoption appears to depend on whether workers possess the capability to effectively interpret and integrate AI outputs and whether their tasks are sufficiently structured to support reliable use. This implies that organizations may benefit from targeting AI deployment toward roles with clearer task structures, investing in digital capability development before scaling AI broadly, and treating training and task redesign as complements to AI adoption rather than as separate initiatives.

Rather than focusing solely on increasing AI usage, organizations may benefit from investing in training and support systems that enhance employees' ability to effectively interpret and apply AI outputs. In addition, structuring tasks in ways that reduce ambiguity may help stabilize the returns to AI usage, particularly for less experienced or less capable users.

More broadly, the results suggest that AI adoption strategies should account for heterogeneity in user capability and task environments, rather than assuming uniform productivity gains across the workforce.

CONCLUSION

This paper examines how the relationship between AI usage and performance depends on worker capability and task structure. Using cross-sectional survey data, the analysis shows that AI usage is generally associated with improvements in performance, but that the magnitude of these improvements varies systematically across individuals and tasks.

The results suggest that AI functions as a capability-contingent amplifier, producing greater performance gains for workers who are better able to effectively use and interpret its outputs. At the same time, task structure shapes the consistency of these gains, with more standardized tasks yielding more stable relationships between AI usage and performance.

These findings highlight the importance of moving beyond average effects in understanding the impact of AI on work. Rather than asking whether AI improves performance, future research should focus on identifying the conditions under which its benefits are maximized and how these benefits are distributed across workers. By emphasizing heterogeneity and complementarity, this study provides a step toward a more nuanced understanding of the role of AI in knowledge work.

A key next step is to test this framework using larger multi-source datasets that combine behavioral measures of AI usage with supervisor-rated or administrative performance outcomes. In particular, future research could incorporate passive digital trace data—such as system logs from AI tools, workflow metadata, or platform usage records—to capture how frequently, intensively, and in what contexts employees engage with AI in their daily work. Such designs would allow more rigorous assessment of the causal and configurational claims that the present pilot is only able to examine indirectly.

APPENDIX

This appendix reports the survey items used in the questionnaire. Table 3 lists the measures used in the main analysis. Table 4 lists additional screening, descriptive, and quality-check items.

Table 3: Main analysis measures

Survey item	**Response scale / coding**
Task standardization	
My core tasks have clear procedures or best practices.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
Criteria for good performance are well-defined for my core tasks.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
My core tasks usually have a clear correct solution.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
My core tasks require substantial judgment because the correct approach is often unclear.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
AI usage intensity	
How often do you use AI assistance tools for work?	1 = Never; 2 = Less than once per month; 3 = 1–3 times per month; 4 = 1–2 times per week; 5 = 3–4 times per week; 6 = Daily or more
Approximately what percentage of your work time involves using AI assistance tools?	Open numeric response (0–100)
AI assistance tools are integrated into my regular workflow.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
Digital capability	
I am comfortable learning and using new digital tools for work.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
Performance	
Compared to others in similar roles, how would you rate your performance over the past 3 months?	1 = Bottom 20%; 2 = Below average; 3 = About average; 4 = Above average; 5 = Top 20%

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Survey item	Response scale / coding
Have you received a formal performance evaluation in your current role within the past 12 months?	1 = Yes; 2 = No; 3 = Not sure
How would you describe the overall outcome of your most recent performance evaluation? (Displayed only if formal evaluation = Yes.)	1 = Well below expectations; 2 = Below expectations; 3 = Meets expectations; 4 = Exceeds expectations; 5 = Far exceeds expectations
Over the past 4 weeks, I consistently met deadlines.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
Over the past 4 weeks, my work quality met or exceeded expectations.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
Over the past 4 weeks, I produced a high volume of output relative to my role.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
AI tools and activities	
Which AI assistance tools do you use for work? (Select all that apply.)	Multiple choice: ChatGPT; Microsoft Copilot; Google Gemini; Claude; Grammarly or similar; Notion AI or similar; Perplexity or similar; Company-provided internal AI tools; Other; I do not use AI assistance tools
For which activities do you use AI assistance tools at work?	Multiple choice: Writing or editing text; Summarizing documents or messages; Brainstorming or ideation; Coding or debugging; Data analysis; Planning or outlining; I do not use AI assistance tools

Table 4: Additional survey items

Survey item	Response scale / coding
Screening and employment	
Are you 18 years of age or older?	1 = Yes; 2 = No
Are you currently employed (including self-employed)?	1 = Yes; 2 = No
You indicated that you are currently employed. Is that correct?	1 = Yes; 2 = No
Job characteristics	
How many hours per week do you typically work in your primary job? (Please enter a number. If your hours vary, give your best estimate.)	Open numeric response

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Survey item	Response scale / coding
Which best describes your current primary job function?	1 = Software / IT / Engineering; 2 = Data / Analytics / Research; 3 = Operations / Supply Chain; 4 = Sales / Account Management; 5 = Marketing / Communications; 6 = Finance / Accounting; 7 = HR / People Operations; 8 = Customer Support / Service; 9 = Legal / Compliance; 10 = Product / Program Management; 11 = Other (specify)
Which best describes your current job level?	1 = Individual contributor; 2 = Team lead / supervisor; 3 = Manager; 4 = Senior manager / director; 5 = Executive; 6 = Other (specify)
How long have you been in your current role? (Decimals are OK, e.g., 0.5.)	Open numeric response
How many years of full-time work experience do you have? (Decimals are OK.)	Open numeric response
In one sentence, what is your primary job?	Open text response
AI perceptions	
AI assistance tools help me complete tasks faster.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
AI assistance tools improve the quality of my work.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
AI assistance tools often create extra work because I need to verify or correct outputs.	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree
Attention and quality checks	
To show that you are paying attention, please select 'True.'	1 = True; 2 = False
Please select 'Agree.'	1 = Strongly disagree; 2 = Disagree; 3 = Neither agree nor disagree; 4 = Agree; 5 = Strongly agree

Note. The main analysis uses the following constructs: task standardization (task_clarity, performance_criteria, solution_clarity, task_judgment [reverse-coded]), AI usage intensity (ai_frequency, ai_percent, ai_integration), and digital capability (digital_comfort). The formal evaluation outcome is based on evaluation_outcome, observed only among respondents who reported receiving a formal evaluation within the past 12 months.

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